

Milestone 1 – Sentiment Analysis Report

Project: Product Sentiment Analysis and Feature Roadmap Prioritization

Course: DA 401 – Minor in Business Analytics

Milestone: M1 – Review Analysis & Sentiment Classification

1. Dataset Overview

1.1 Business Context

FitLife is a Health & Fitness mobile app with approximately 50,000 monthly active users and an average rating of 4.1 on the app store. Despite strong acquisition, growth has stalled and recent competitors with higher ratings (≈ 4.5) have increased competitive pressure. This analysis evaluates competitor review sentiment to identify improvement opportunities.

1.2 Data Source and Filtering

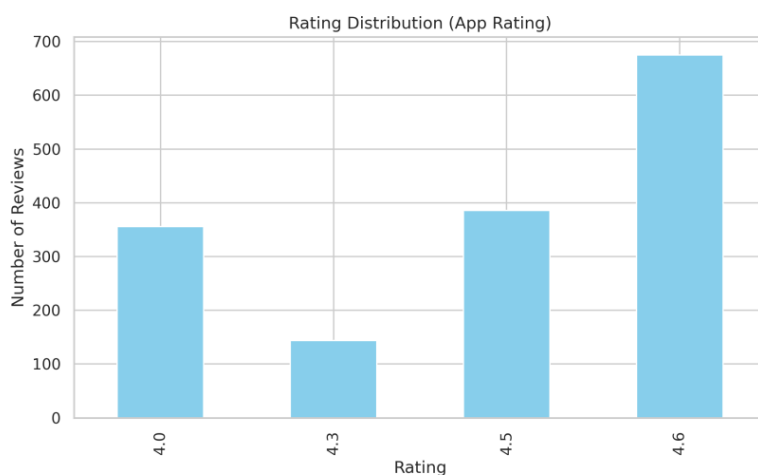
- Source: Google Play Store Apps and Reviews dataset (Health & Fitness category).
- Scope: Top apps by review volume within Health & Fitness category.
- Apps with fewer than 100 reviews were excluded.
- App metadata and reviews were merged on the App column.
- Final dataset used: M1_Reviews_Enriched_For_M2.csv.

1.3 Final Analysis Sample

Number of apps analyzed: 7

Number of clean reviews: 1561

Time dimension: Cross-sectional (review dates not used for time-series analysis).



2. Data Cleaning and Preparation

2.1 Cleaning Steps

- Dropped rows where Translated_Review was null, empty, or 'nan'.
- Removed reviews shorter than 10 characters.
- Removed exact duplicate reviews (same App and Translated_Review).
- Ensured Rating stored as numeric value.
- Computed Review_Length feature.
- Applied VADER SentimentIntensityAnalyzer and stored results in Sentiment_VADER column.

2.2 Cleaning Outcomes

Final clean review count: 1561

3. Descriptive Analysis of Ratings and Reviews

3.1 Rating Distribution

4.0: 356 (22.81%)

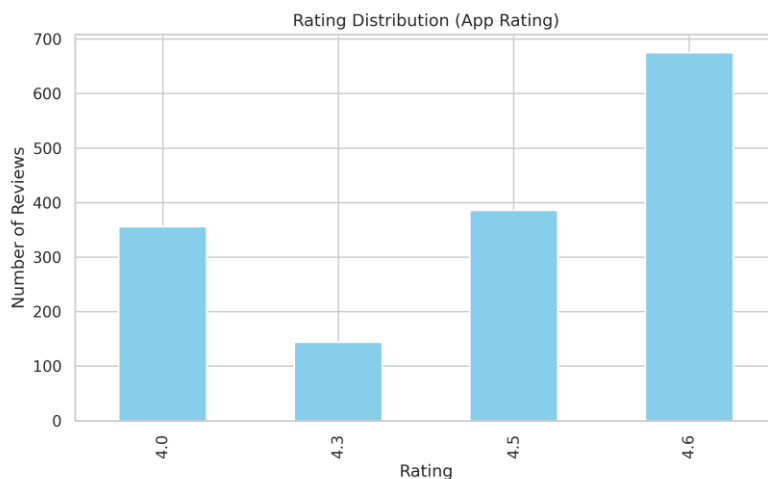
4.3: 144 (9.22%)

4.5: 386 (24.73%)

4.6: 675 (43.24%)

Overall Average Rating: 4.4108

Ratings are concentrated in higher buckets (4.5 and 4.6), indicating generally strong user satisfaction, though a meaningful portion exists in lower rating buckets.



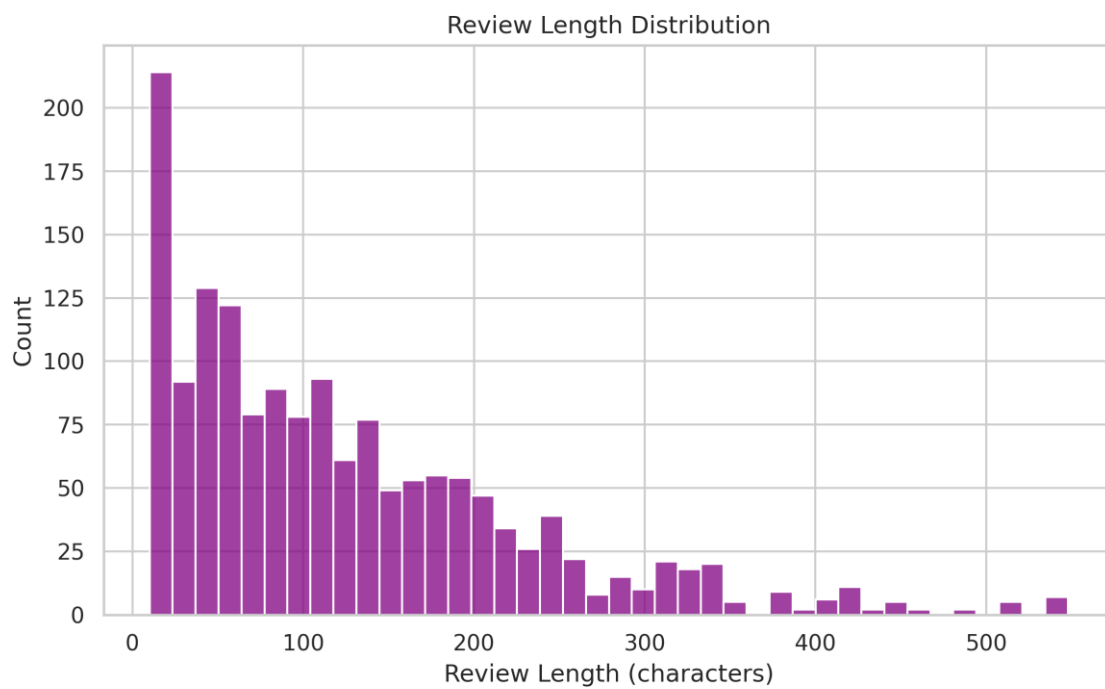
3.2 Review Length Analysis

Average review length: 125.35 characters

Median review length: 101 characters

Min–Max review length: 10 – 548 characters

Higher rating bucket (4.6) reviews average 142.13 characters, while lower buckets (4.0 & 4.3 combined) average approximately 103.09 characters. This suggests longer reviews often contain richer qualitative feedback.

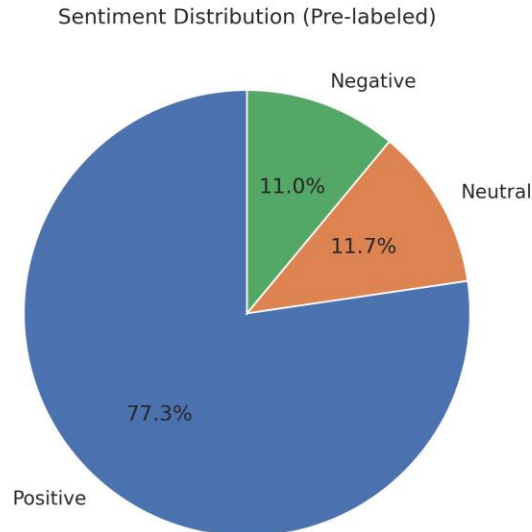


3.3 Pre-Labeled Sentiment Distribution

Positive: 1207 (77.32%)

Neutral: 182 (11.66%)

Negative: 172 (11.02%)



4. Sentiment Analysis Using VADER

4.1 Methodology

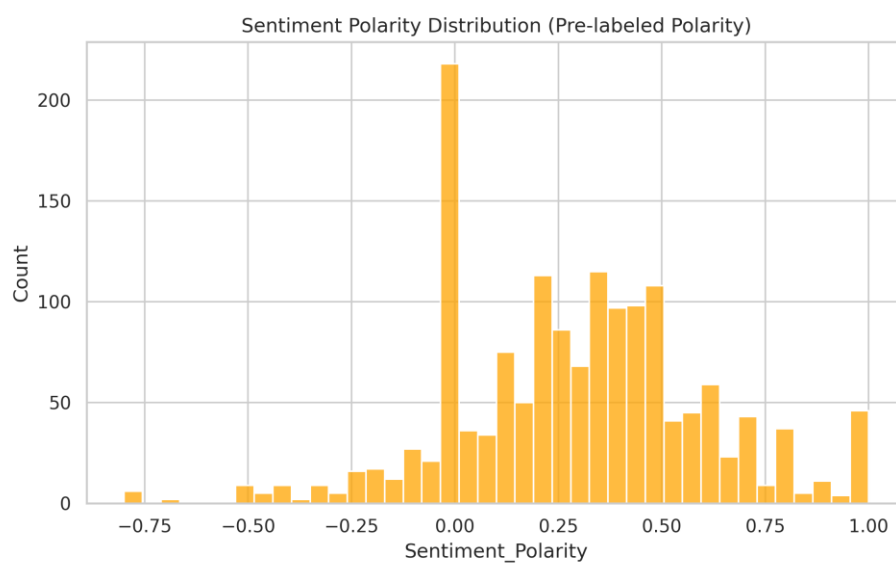
VADER SentimentIntensityAnalyzer was applied to each review. Classification rules: Positive ($\text{compound} \geq 0.05$), Negative ($\text{compound} \leq -0.05$), Neutral ($-0.05 < \text{compound} < 0.05$).

4.2 VADER Sentiment Distribution

Positive: 1220 (78.16%)

Neutral: 152 (9.74%)

Negative: 189 (12.11%)



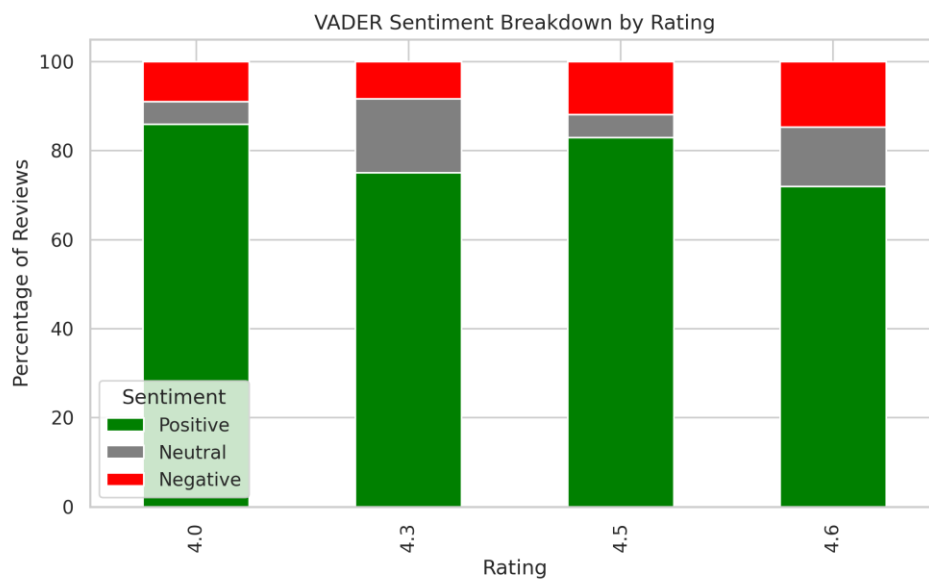
4.3 Validation vs Pre-Labeled Sentiment

Overall agreement: 79.69%

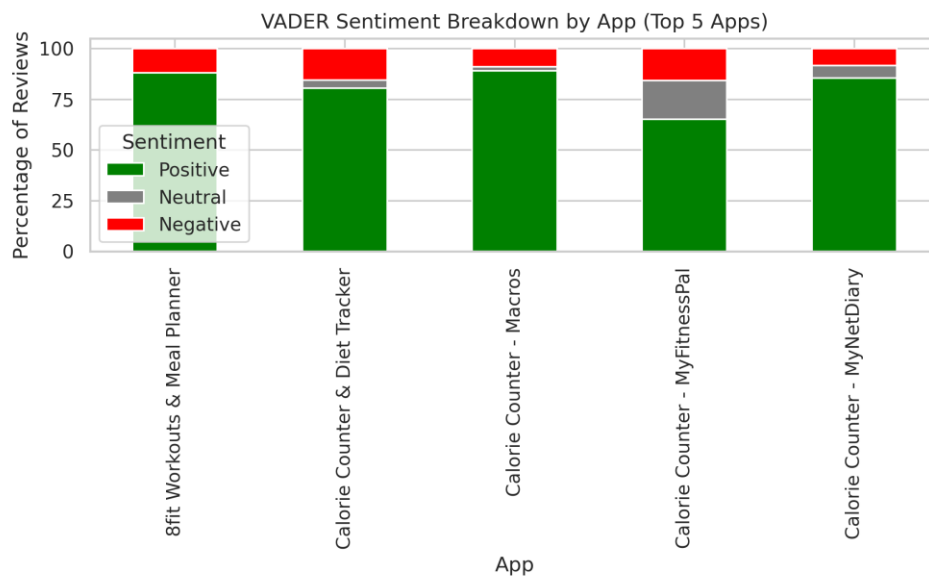
The confusion matrix shows strong alignment between VADER predictions and pre-labeled sentiment, confirming reliability for downstream feature-level analysis.

5. Relationship Between Ratings and Sentiment

For rating bucket 4.6, 72.00% are Positive and 14.67% are Negative. For rating bucket 4.0, 85.96% are Positive and 8.99% are Negative. Rating 4.3 shows a relatively higher Neutral share (16.67%).



Average sentiment polarity varies across rating buckets, showing alignment between numerical ratings and text polarity.



6. App Level Sentiment Insights

Calorie Counter - MyNetDiary: 0.3999

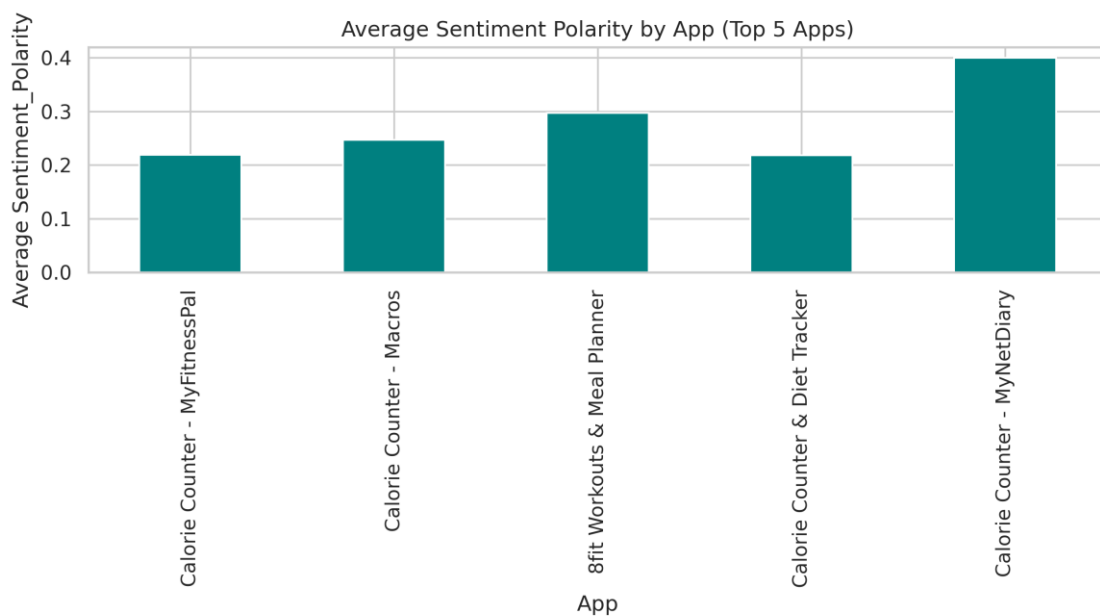
8fit Workouts & Meal Planner: 0.2972

Calorie Counter - Macros: 0.2475

Calorie Counter - MyFitnessPal: 0.2191

Calorie Counter & Diet Tracker: 0.2183

Calorie Counter - MyNetDiary shows the highest average sentiment polarity, indicating stronger user satisfaction relative to competitors.



7. Key Insights and Implications

1. Overall sentiment is strongly positive but contains actionable negative feedback (~11–12%).
2. VADER is a reliable sentiment signal (~79.7% agreement).
3. Longer reviews provide richer qualitative insights.
4. App-level polarity differences highlight benchmarking opportunities.
5. With a cleaned dataset and validated sentiment scores, Milestone 2 will focus on topic extraction and feature-level analysis to identify drivers of sentiment.